GrappStar build guide

One guide from parts to first mission: print, wire, flash, calibrate, test. Every hands-on step is in order. Every wiring step starts with the battery unplugged.

Built 2026-11-03 10:55 Sydney time from commit `acb5b4c`.

How to use it: read section 1 and 2 first, then work through the "Order of work" list in section 1. Where a section says "From `some/file.md`", that text comes straight from the repo, so it is always the latest version. Re-build the guide after any change: `python3 docs/build/make_guide.py`.

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1. What the system is

GrappStar is a rover that can pull itself out of trouble. A launcher on the rover fires a small dart (KEDGE) on a Dyneema tether. The dart lands, screws itself into the soil, and the rover winches against it. Then the dart lets go and is reeled home.

Three parts:

- **Rover.** Jetson Orin Nano Super on the rover's 3S LiPo. It runs the mission, the operator web page, and the vision (rover camera Razer Kiyo X, and the dart's video through an RC832 receiver and a DIGITNOW USB capture).
- **Base station** (on the rover). ONE 12 V wiper motor drives everything. A sliding dog collar on the main shaft couples the wiper to the LAUNCH train, to nothing (NEUTRAL), or to the REEL train. A small N20 motor turns a selector drum (like a motorcycle gear drum) that moves the collar and also lays and removes the capstan wraps.
  - LAUNCH, wiper forward: draws the spring carriage back, fires at the end of the stroke, resets.
  - LAUNCH, wiper backward: tilts the rail up and down to aim. A self-locking worm holds the angle.
  - REEL, wiper forward: the capstan winds in; the storage reel takes up the slack through a slip clutch.
  - Payout at launch: the line skips the capstan and pulls off the free-spinning storage reel against a light brake (0.5 to 1 N).
  - Any fault: wiper off, drum to NEUTRAL. There is a hand lever on the drum shaft end.
  - Brain: an ESP32-S3 DevKitC on USB from the Jetson, with an AS5600 angle sensor on the main shaft and printed contact switches (an M3 screw head touching an M3 nut).
- **Dart (KEDGE v30).** 42 mm x 152 mm, its own XIAO ESP32S3, 2S 18350 pack, WiFi access point. After landing a kick leg swings over the nose and plants a foot auger, then two drill pods swing down and screw in. A nose camera sends analog video at 5.8 GHz.

How they talk: the Jetson talks to the base over USB serial (JSON lines, `firmware/base/PROTOCOL.md`) and to the dart over WiFi (HTTP at 192.168.4.1, `firmware/dart/API.md`). Video is analog 5.8 GHz into the USB capture.

A mission: pick a target on the rover camera view, the Jetson looks up the pitch in the ballistics table, the base aims, cocks and fires on your confirmation, the dart video is analysed for the landing zone, the dart anchors, the base wraps the capstan and tows, the dart releases, the base reels it home.

Order of work

1. Measure the wiper motor and the springs (section 14, groups A and B). Put the numbers in `base/spec/params.py` and re-run the base pipeline. The base parts that touch the motor depend on them.
2. Confirm the parts box (section 14, group C). The only purchase is 608 bearings.
3. Print the dart fit coupon, then the dart plates (section 4). The dart does not depend on the measurements.
4. Flash the dart and build it (sections 6 and 8, and the dart run sheet in the appendix).
5. Print the base (section 4) and assemble it (section 5).
6. Wire the base and the rover (section 7). Flash the base ESP32 (section 8). Set up the Jetson (section 9).
7. Calibrate (section 10) and bench test (section 11).
8. Run the staged tests, bench to field (section 12).

2. Safety

Read this before any hands-on step.

Electrical

- **Battery unplugged before you touch, add or move any wire.** On the dart that means the power loop plug out. The first XIAO died from wiring with the battery live.
- Battery unplugged before you plug any board into the Mac by USB.
- Check for shorts with the meter on beep before the first power-on of anything you wired: red to black must not beep.
- The 3S LiPo can deliver over 100 A into a short. Keep the 20 A and 10 A fuses in their leads. Never bypass them.
- LiPo care: charge on the iMax B6 in balance mode, on a non-flammable surface, never unattended. Stop using the pack at 3.3 V per cell under load (9.9 V). A puffed pack goes in a LiPo bag and to battery recycling.
- 18350 cells: never let both cells touch metal at the same time. Charge out of the dart in the XTAR, or through the tail balance socket with the loop plug out.
- Never connect the DRV8833 motor supply (VM) to the LiPo directly: 10.8 V maximum. It runs from the 9.0 V buck.
- The dart VTX must have its antenna on before it is powered.

Mechanical

- The wiper motor is strong (about 15 N m) and self-locking. Hands out of the gears, drum and rail while the battery is plugged in. Pull the battery plug to stop everything.
- The launch springs store energy. Cocked carriage = loaded. Never put a hand in front of the rail or the carriage. To unload a cocked base without firing, see troubleshooting.
- From the first spring test on: safety glasses, nobody in front of the rail, the area clear for 30 m.
- Augers and drills are sharp and turn with force. Hold the dart by the hull, never near a pod or the foot.
- Soldering: iron on its stand, fan on for fumes, 3 s at most per joint near PLA. Heat gun on low, no lighter near PLA.

3. Parts

Everything below is already in hand except the 608 bearings. Section 14 group C lists the items to confirm by looking in the box.

The only purchase: 608 bearings

- Search: "608ZZ bearing 8x22x7" (or "608-2RS", sealed). Buy 8 to 10. Skateboard bearings are 608s and fine.
- Specs: 8 mm bore, 22 mm outside, 7 mm wide, rated well over 10,000 rpm.
- Uses: storage reel (2), capstan or wrap arm (2), rollers and spares. Axles: M8 bolt or 8 mm rod (check the box).

Base station and rover

Part	Qty	Note
12 V wiper motor, self-locking worm	1	Rebuilt to 2 wires, park linkage removed. Measure it first (section 14 A)
BTS7960 motor driver module	1	Drives the wiper
ESP32-S3 DevKitC-1-N8	1	Base brain, USB to the Jetson
Adafruit AS5600 (STEMMA QT) + diametric magnet	1	Main shaft angle
DRV8833 carrier	1	DRV8833 #3, freed from the dart by the new dart pin map
MP1584 buck	1	MP1584 #2 from the dart build, set to 9.0 V
N20 right-angle worm motor	1	Selector drum (count your N20s: section 14 C3)
CSK20PP one-way bearings (20 x 47 x 14)	2	#1 spring cock and fire, #2 capstan backstop
Century 16 x 165.1 x 1.4 mm extension springs	2	Launch springs (Bunnings 0487961). Measure them (section 14 B)
608 bearings	8 to 10	The purchase
Dyneema braid 150 lb, 1 mm	20 m	Tether
100 kg load cell + HX711	1	Optional line tension
3S LiPo 5200 mAh (Zeee), iMax B6 charger	1	Rover battery
XT60 lead #1 (10 A fuse), lead #2 (20 A fuse), spare XT60s, fuse holder		Lead #2 feeds the wiper; build the Y harness from spares
Barrel plug lead 5.5 x 2.5 mm	1	Jetson from the LiPo
Jetson Orin Nano Super	1	
Lockmaner RC832 receiver, DIGITNOW USB capture, RCA lead	1 each	Dart video in
Razer Kiyo X	1	Rover camera
Resistors 10 k x 3, 1 k, 680 R, 100 k, 22 k		Pull-downs and dividers (firmware/base/WIRING.md)
M3 screws, M3 nuts, M3 brass heat-set inserts		Frame joints and printed contacts
Dupont jumpers, 14 / 18 / 22 / 26 AWG wire, heat shrink		

Dart (KEDGE v30)

The full dart list with quantities is in the dart run sheet, section 2 (appendix). In short: XIAO ESP32S3, 2 x DRV8833 (the third goes to the base), INA219, BNO085, MP1584 #1 (5.04 V), OTOTEC 2S BMS, 2 x 18350 cells, 3 worm N20s (kick, deploy, foot), 2 drill gearmotors, 5 x F693ZZ, Caddx Ant, Zeus VTX and antenna, JST-XH sockets, Schottky diode, capacitor, M3 hardware, 15 g nose ballast.

Tools

Soldering iron and solder, flux, multimeter, wire strippers, flush cutters, small flat screwdriver (MP1584 trimmer), heat shrink and heat gun, caliper, hex keys, tweezers, kitchen scale, luggage scale, tape measure, safety glasses, 5-minute epoxy, threadlocker, zip ties, foam tape.

4. Printing

Printer: Bambu X2D (bed 256 x 256 x 260 mm). PLA for everything except loaded gears and keyed shafts, which are PPA-CF (the print list says which).

Order: dart fit coupon first (20 minutes, checks your printer's hole sizes), then the dart plates, then the base once the wiper and spring numbers are in (section 1, order of work).

Dart prints

Dart plates (in `kedge/pack_v30/`, also inside `kedge/dist_v30/KEDGE_v30_build_pack.zip` under `print/`):

- `kedge/pack_v30/plate_A_hull.3mf`
- `kedge/pack_v30/plate_B_legs_tray.3mf`
- `kedge/pack_v30/fit_coupon.stl` (print this first, 20 minutes)

Dart part	Print pose	Size on bed mm	Volume cm3
hull_top	split face down (on the bed), saddle up	121 x 50 x 35	17.41
hull_bottom	split face down, belly up	121 x 42 x 21	11.5
nose	tip down on the bed (flat Ø22), spigot up	42 x 42 x 27	8.25
tail_cap	end face down, spigot up (no supports)	42 x 42 x 15	7.13
tray	keel down on the bed; discs stand up, decks bridge 26 mm between discs	109 x 39 x 36	15.98
kick_leg	on its side (pole axis horizontal, hub face down)	119 x 15 x 15	8.7
drill_leg_L	upright, motor pocket open to the top (as installed folded)	46 x 15 x 13	3.37
drill_leg_R	upright, motor pocket open to the top (as installed folded)	46 x 15 x 13	3.37
drill_leg_L_auger	base down, nut slot sideways	17 x 17 x 73	5.5
drill_leg_R_auger	base down, nut slot sideways	17 x 17 x 73	5.5
foot_auger	base down	17 x 17 x 40	3.06
coupler_L	standing on its socket end	7 x 6 x 36	1.24
coupler_R	standing on its socket end	7 x 6 x 27	0.9

Settings: PLA, 0.4 nozzle, 0.16 mm layers, 4 walls, 40% gyroid infill, 5 top and bottom layers, supports off, brim on for the augers and couplers. Hole compensation from the fit coupon (run sheet section 4).

Base prints

The base print pack comes from the base pipeline (`base/pack`). It holds every part already in print orientation, the plates as 3MF, and the material for each part.

> *NOT IN THE REPO YET:* `base/pack` *(written by cad-infra (print pack)). Re-run* `python3 docs/build/make_guide.py` *after it lands and this section fills in.*

Before printing the base:

1. The wiper and spring numbers are in params.py and the pipeline was re-run (the audit said CLEAN).
2. Print one test part from the pack in PPA-CF and check a 608 drops into its seat with a firm thumb press, and an M3 heat-set insert sets straight. Adjust hole compensation the same way as the dart coupon.
3. PPA-CF: dry the spool first (it absorbs water), hardened nozzle, settings from the Bambu PPA-CF profile.

5. Base assembly

The assembly steps come from the base pipeline: every step is generated from the same model that passed the collision audit, with a picture per step. They are in `base/guide` (PDF, also attached as an appendix) and `base/out`.

> *NOT IN THE REPO YET:* `base/guide/README.md` *(written by cad-infra (assembly steps)). Re-run* `python3 docs/build/make_guide.py` *after it lands and this section fills in.*

General rules for the base:

- Heat-set inserts: soldering iron at about 220 C with the insert tip, press straight in, let it cool before loading.
- Bearings: press in square with a flat block, never hammer on the inner race.
- Keyed shafts: check the key slides in the sprag (CSK20PP) keyway by hand before final assembly.
- Turn every train by hand (battery unplugged) after each step: it must turn without binding before the next step.
- Leave the springs off until test stage 4 (section 12).

6. Dart build

Build the dart with the KEDGE v30 run sheet (appendix at the end of this guide, also `kedge/pack_v30/KEDGE_v30_run_sheet.pdf`). It is a complete, step by step, timed build day: fit coupon, plates, board prep, harness, power-up tests, motor directions, mechanical assembly, function test.

One change from the run sheet: the dart pin map. The run sheet uses three DRV8833 boards. The dart now uses two, so the third can drive the base selector drum:

- DRV#1: deploy and kick (unchanged).
- DRV#2 channel A: BOTH drill motors (0.7 A stall together, fine on one 1.2 A channel), right drill motor leads swapped as before so the pair counter-rotates.
- DRV#2 channel B: the foot N20.
- No DRV#3 in the dart: skip every run sheet wire that goes to DRV#3, and do not bridge AIN1 to BIN1 on DRV#2.

Where the run sheet wiring (its sections 7 and 9 and its wiring picture) differs from `firmware/dart/WIRING.md` below, follow WIRING.md. Use the firmware in `firmware/dart/`, not the copy inside the v30 pack. Ignore the run sheet's `firmware/reel` sketch: the base firmware in `firmware/base/` replaces it.

The rest of the run sheet stands: the fit coupon, both plates, the 5.04 V buck, the Schottky diode, the capacitor polarity, the battery contacts, the assembly order and the function test.

Dart wiring (new pin map)

> *NOT IN THE REPO YET:* `firmware/dart/WIRING.md` *(written by fw-dart). Re-run* `python3 docs/build/make_guide.py` *after it lands and this section fills in.*

Dart bench tests

The run sheet sections 7.5, 7.6, 8d step 6 and 10 are the dart bench checks: power-up readings, motor directions, battery power-up, then the function test in damp soil.

7. Electronics and wiring (base and rover)

Battery unplugged for every step in this section. Build in this order: the power harness (Y lead, fuses), the base electronics bay, the sensors and contacts, the Jetson and video. Then do the first power-on sequence (7.1 section 4) before anything is connected to a motor.

7.1 The whole system

From `docs/build/SYSTEM_ELECTRICAL.md`*:*

GrappStar system electrical (rover, base and dart)

Diagram: `system_electrical.svg` / `system_electrical.png` (source `system_electrical.dot`). The ESP32 pin table is owned by `firmware/base/WIRING.md` and `firmware/base/grappstar_base/pins.h`; this page shows how the whole rover is powered and connected. If this page and WIRING.md ever disagree on a pin, WIRING.md wins.

1. What is on the rover

Block	Part	Powered from
Battery	Zeee 3S LiPo 5200 mAh, XT60 (9.9 V empty in use, 11.1 V nominal, 12.6 V full)	
Wiper branch	XT60 lead #2 with the 20 A fuse, BTS7960 module, 12 V wiper motor (2 wires, park switch taped off)	LiPo
Drum branch	MP1584 #2 buck set to 9.0 V, DRV8833 #3 (the one freed from the dart), selector drum N20 worm	BTS7960 B+ / B- terminals, after the 20 A fuse
Base brain	ESP32-S3 DevKitC-1-N8	Jetson USB (the port marked USB, not UART)
Sensors on the ESP32	Adafruit AS5600 (STEMMA QT, I2C 0x36), printed contacts (M3 screw head on an M3 nut), optional HX711 + 100 kg load cell, battery divider	ESP32 3V3
Jetson	Orin Nano Super dev kit	XT60 lead #1 (10 A fuse) to a 5.5 x 2.5 mm barrel pigtail
Video in	Lockmaner RC832 5.8 GHz receiver, AV (yellow RCA) out to the DIGITNOW USB capture, which plugs into the Jetson	XT60 lead #1 branch (12 V class input)
Rover camera	Razer Kiyo X	Jetson USB
Dart	KEDGE: XIAO ESP32S3, 2 x DRV8833, INA219, BNO085, Caddx Ant, Zeus VTX	its own 2S 18350 pack. No wire to the rover: the tether is Dyneema and carries no power

2. Power tree

```
3S LiPo (XT60)
  -> XT60 Y harness (one male in, two female out; made from your spare XT60s, 14 AWG)
       -> lead #2, 20 A fuse, 14 AWG -> BTS7960 B+ / B-  -> wiper motor (BTS7960 M+ / M-, 14 AWG)
                                      \-> 22 AWG from the B+ / B- screws -> MP1584 #2 IN (9.0 V out)
                                            -> DRV8833 #3 VM / GND -> drum N20 (AOUT1 / AOUT2)
       -> lead #1, 10 A fuse, 18 AWG -> barrel pigtail -> Jetson DC jack
                                      \-> 22 AWG -> RC832 power in
Jetson USB -> ESP32-S3 DevKitC (5 V from USB; the board makes its own 3V3)
                ESP32 3V3 -> BTS7960 VCC, DRV8833 SLP (via GPIO), AS5600, HX711
Jetson USB -> DIGITNOW capture <- RC832 AV out
Jetson USB -> Kiyo X
```

Why it is laid out this way:

- The DRV8833 motor supply (VM) is 10.8 V maximum. The LiPo is 12.6 V full, so the drum N20 runs from MP1584 #2 set to 9.0 V. MP1584 #2 is the spare from the dart build (it was the 8.0 V bench supply there): reset it to 9.0 V after the dart bench test.
- The buck taps the BTS7960 screw terminals, so both motor branches sit behind the 20 A fuse and the Jetson branch stays separate. A wiper stall that blows the 20 A fuse does not reboot the Jetson.
- The ESP32 takes power only from USB. It never connects to the LiPo directly. Its ground meets battery minus through the BTS7960 and DRV8833 GND wires (needed so the logic signals have a reference).

3. Wires and connectors

Run	From	To	Gauge	Connector	Note
P1	LiPo	Y harness	14 AWG	XT60	Make the Y from spare XT60s; solder, heat shrink each joint
P2	Y harness	BTS7960 B+ / B-	14 AWG	XT60 lead #2 (20 A fuse), screw terminals at the BTS7960	Red to B+, black to B-. 10 A blew on wiper inrush before: keep 20 A

Run	From	To	Gauge	Connector	Note
P3	BTS7960 M+ / M-	wiper motor	14 AWG	screw terminals; spade or solder at the motor	Red wire (+) and black (-) of the rebuilt 2-wire motor
P4	BTS7960 B+ / B-	MP1584 #2 IN+ / IN-	22 AWG	under the same screws	Set 9.0 V with nothing on the output first
P5	MP1584 #2 OUT+ / OUT-	DRV8833 #3 VM / GND	22 AWG	solder	Check 9.0 V on the meter before you connect
P6	DRV8833 #3 AOUT1 / AOUT2	drum N20	26 AWG	solder or JST	Swap the pair if the drum turns the wrong way (see calibration)
P7	Y harness	Jetson DC jack	18 AWG	XT60 lead #1 (10 A fuse), 5.5 x 2.5 mm barrel plug (same as the plug on the Jetson supply), centre positive	Check polarity with the meter before the first plug-in
P8	lead #1 (after the fuse)	RC832 power in	22 AWG	the RC832 power lead	Read the RC832 label for its input range first (see MEASURE_LIST)
S1	Jetson USB-A	ESP32-S3 USB port (native, marked USB)		USB-C or micro-USB cable to suit the board	Data cable, not charge-only
S2	Jetson USB-A	DIGITNOW capture		its own USB plug	
S3	RC832 AV out	DIGITNOW video in		yellow RCA	Video only; audio not used
S4	Jetson USB-A	Kiyo X		its own USB plug	
L1	ESP32 pins	BTS7960 RPWM, LPWM, R_EN, L_EN, IS, VCC, GND	26 to 28 AWG	Dupont jumpers	Pins in firmware/base/WIRING.md
L2	ESP32 pins	DRV8833 #3 AIN1, AIN2, SLP, GND (nFAULT if present)	26 to 28 AWG	Dupont	
L3	ESP32 SDA / SCL / 3V3 / GND	AS5600		STEMMA QT to Dupont cable	Magnet 1 to 3 mm from the chip
L4	ESP32 pins	each printed contact	26 AWG twisted pair	Dupont at the ESP32, M3 ring or solder at the contact	One wire to the pin, the other to GND
L5	ESP32 pins	HX711 DT, SCK, VCC, GND (optional)	26 AWG	Dupont	Load cell 4 wires into HX711 E+, E-, A+, A-
L6	BTS7960 B+	ESP32 battery pin through the 100 k / 22 k divider	26 AWG	solder at the divider	12.6 V reads 2.27 V at the pin

Gauges assume runs under 0.5 m. 14 AWG carries the 18 A wiper stall; 18 AWG is fine for the Jetson (about 2.5 A at 10 V); 22 AWG is fine for the drum branch (N20 stall about 0.2 A) and the RC832 (about 0.2 A).

4. First power-on sequence (do it in this order, once, after any rewiring)

Battery unplugged for steps 1 to 4. Have the multimeter ready.

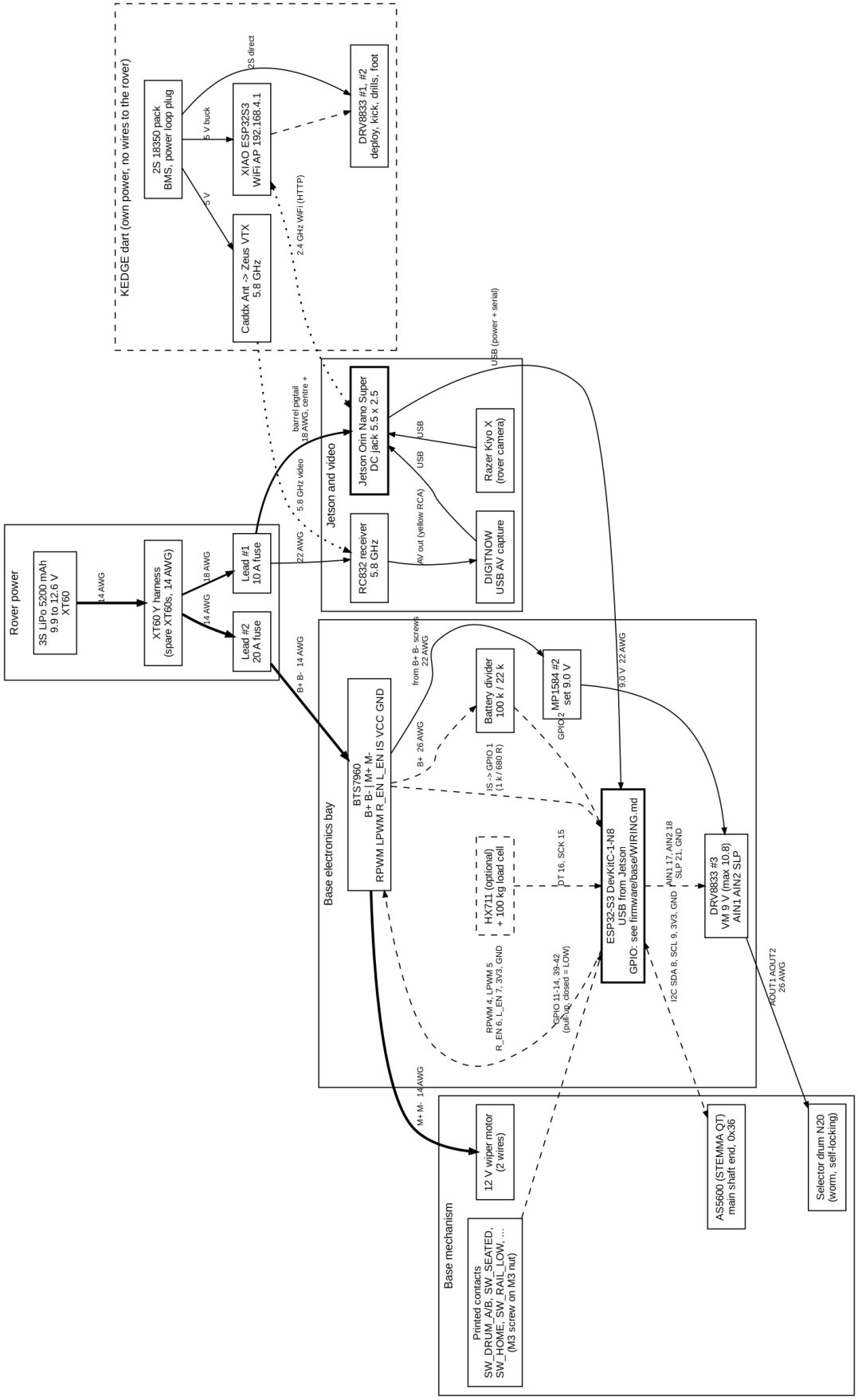
1. Battery unplugged. Pull both fuses. Meter on beep: touch red and black of every branch (P2 at the BTS7960, P4, P5, P7 at the barrel plug, P8). Nothing may beep. A beep is a short: find it before going on.
2. Battery unplugged. Meter on beep: barrel plug centre pin to the red wire of lead #1 must beep (centre positive).
3. Battery unplugged. DRV8833 #3 VM wire disconnected from MP1584 #2 OUT. ESP32 USB cable unplugged.
4. Battery unplugged. Meter on DC volts on the LiPo balance lead: each cell 3.7 to 4.2 V.
5. Plug the battery in. Fit the 20 A fuse only. Meter on MP1584 #2 OUT: turn the brass screw until it reads 9.0 V. The wiper must not move (the BTS7960 enables are held low by their pull-downs). If it moves: unplug the battery and check R_EN / L_EN.
6. Unplug the battery. Solder or plug the DRV8833 VM wire (P5) to MP1584 #2 OUT.
7. Plug the battery in. Fit the 10 A fuse. The Jetson boots (green LED). The RC832 screen or LED lights.
8. Plug the ESP32 USB cable into the Jetson. The ESP32 power LED lights. Nothing moves (DRV8833 SLP has a pull-down and stays off until the firmware enables it).
9. On the Jetson: `python3 -m grappstar.cli ping` (see jetson/README.md) or a serial terminal: `{"id":1,"cmd":"status"}`. Check `vbat` in telemetry matches the meter within 0.3 V.

Unplug order (every time): stop the motors (`estop`), pull the battery, then anything else.

5. Rules

- Battery unplugged before you touch, add or move any wire. This killed the first XIAO.

- Battery unplugged before you plug the ESP32 into the Mac. Flash from the Mac with the ESP32 out of the rover harness, or flash from the Jetson (firmware/base/README.md).
- Stop using the LiPo at 3.3 V per cell under load (9.9 V); the firmware reports `vbat` and the Jetson warns.
- Never connect the DRV8833 VM to the LiPo directly (10.8 V maximum; the LiPo is 12.6 V full).
- Never put 3V3 on a DRV8833 pad called VIN, VCC or VM: on these carriers that is the motor supply. 3V3 only reaches the SLP / nSLEEP / EEP pad, and here it comes from an ESP32 GPIO.
- Dart: power loop plug out whenever a wire moves or USB is plugged in (the dart run sheet rule).
- The VTX on the dart must have its antenna on before it is powered.



Whole system electrical diagram (bold = battery power, dashed = signal, dotted = radio)

7.2 Base ESP32 wiring (pin by pin)

NOT IN THE REPO YET: `firmware/base/WIRING.md` *(written by fw-base). Re-run* `python3 docs/build/make_guide.py` *after it lands and this section fills in.*

8. Flashing both ESP32s

Rules for both boards: battery unplugged (dart: power loop plug out) before the USB cable goes in. Use a data USB cable. Flash the dart XIAO before it goes into the hull (run sheet section 5).

8.1 Base: ESP32-S3 DevKitC-1-N8

> *NOT IN THE REPO YET:* `firmware/base/README.md` *(written by fw-base). Re-run* `python3 docs/build/make_guide.py` *after it lands and this section fills in.*

8.2 Dart: XIAO ESP32S3

Arduino IDE: board XIAO_ESP32S3, Tools > USB CDC On Boot: Enabled (the sketch refuses to compile otherwise). Libraries: Adafruit INA219, Adafruit BNO08x. Open `firmware/dart/kedge_dart/kedge_dart.ino`. After the first USB upload you can upload over WiFi (port "kedge").

> *NOT IN THE REPO YET:* `firmware/dart/README.md` *(written by fw-dart). Re-run* `python3 docs/build/make_guide.py` *after it lands and this section fills in.*

9. Jetson setup

The Jetson already runs JetPack 6.2 (Ubuntu 22.04, Python 3.10) from the SD card with WiFi and SSH. Keep numpy below 2 and do not pip install opencv-python (both broke things in August).

> *NOT IN THE REPO YET:* `jetson/README.md` *(written by jetson). Re-run* `python3 docs/build/make_guide.py` *after it lands and this section fills in.*

10. Calibration (base)

Do it after the base is assembled and wired, springs off, before test stage 3. The values are stored on the ESP32.

> *NOT IN THE REPO YET:* `firmware/base/CALIBRATION.md` *(written by fw-base). Re-run* `python3 docs/build/make_guide.py` *after it lands and this section fills in.*

11. Bench tests

11.1 Base

NOT IN THE REPO YET: `firmware/base/BENCH_TEST.md` *(written by fw-base). Re-run* `python3 docs/build/make_guide.py` *after it lands and this section fills in.*

11.2 Dart

The dart bench checks are in the run sheet (appendix): section 7.5 (power-up readings), 7.6 (motor directions), 8d step 6 (battery power-up) and section 10 (function test and calibration in damp soil). With the new pin map the foot is on DRV#2 channel B: key 4 must still turn the foot shaft clockwise seen from the shaft end.

12. First tests, bench to field

From `docs/build/TEST_PLAN.md`*:*

First tests: bench to field

Do the stages in order. Do not skip ahead when a stage fails: fix it, then repeat the stage. Write the date and the result of each stage in a notebook (or `docs/build/test_log.md`).

How to send base commands: the operator page on the Jetson (jetson/README.md) has buttons for each of these. From a terminal the same commands go over the serial link as JSON lines (firmware/base/PROTOCOL.md), for example `{"id":1,"cmd":"jog","motor":"wiper","dir":1,"ms":300}`. `{"id":2,"cmd":"estop"}` stops everything and is always safe to send.

Safety for every stage:

- Battery unplugged before you touch any wire.
- Hands out of the gears, the drum and the rail whenever the battery is plugged in.
- From stage 5 on: nobody in front of the rail, safety glasses on, the launch area clear for 30 m.
- Keep a hand near the battery plug. Pulling it is the real emergency stop.

Stage 0. Power-on (no motors attached)

Follow SYSTEM_ELECTRICAL.md section 4, the first power-on sequence. Both motor leads disconnected from the drivers.

Pass: no beeps on the short check, MP1584 #2 reads 9.0 V, the Jetson boots, `status` replies, `vbat` within 0.3 V of the meter, `fault` is null. If it fails: a beep means a short (check the Y harness and the BTS7960 screws first). No reply: check the USB cable is a data cable and in the ESP32 port marked USB; see troubleshooting in the build guide.

Stage 1. Each motor alone, out of the base

Battery unplugged. Connect the wiper motor to BTS7960 M+ / M- and clamp the motor to the bench. Connect the drum N20 to DRV8833 #3 AOUT1 / AOUT2, loose on the bench. Battery in.

1. `jog` wiper dir 1 for 300 ms, then dir -1. Mark which way the shaft turns for dir 1.
2. `jog` drum dir 1 for 500 ms, then dir -1. Mark the shaft direction.
3. Watch `amps` during each wiper jog.

Pass: each motor turns both ways, stops when the jog ends, wiper free current about 1 to 3 A (A6 in MEASURE_LIST.md), no fault. If it fails: no movement on the wiper: check R_EN / L_EN wiring and the 20 A fuse. Wiper turns only one way: one of RPWM / LPWM is not connected. Drum does nothing: check SLP wiring and 9.0 V on VM. Wrong direction against CALIBRATION.md: swap that motor's two leads (battery unplugged).

Stage 2. Sensors alone

1. Turn the main shaft (or the magnet) by hand: `wiper_deg` changes smoothly, no jumps, and goes up when the shaft turns the FORWARD way.
2. Touch each printed contact closed with a finger: its entry in `sw` changes and changes back when released.
3. Load cell (if fitted): press it, `tension_n` rises.

Pass: every sensor responds and returns. If it fails: AS5600 not found: check SDA / SCL are not swapped and the STEMMA cable is pushed in; angle jumps: magnet too far (keep 1 to 3 mm) or off centre. Contact never changes: check the GND wire and that the screw head really touches the nut.

Stage 3. Base assembled, dry cycles, NO springs, no line

Springs off, no tether, no dart. Run CALIBRATION.md first (zeros and drum positions).

1. `mode` LAUNCH, NEUTRAL, REEL, NEUTRAL, LAUNCH: 20 full rounds. Each change ends with the right `drum_pos`.
2. `aim` 15, 30, 45, then the minimum and maximum pitch: 5 rounds. Rail reaches each angle within 1 degree (`pitch_deg`) and holds with the motor off.
3. Cock and fire with no springs: `arm`, then `fire` with the token. The carriage travels back, releases, and the big gear re-engages: 10 cycles.
4. REEL mode: `wrap`, then `unwrap`: 10 cycles. Then `reel` at duty 0.3 for 3 s with no line.

5. Send `estop` in the middle of each kind of move once. The wiper stops at once and the drum goes to NEUTRAL.
6. Fault test: unplug the AS5600 cable while idle (battery can stay in for this one, nothing moves). The base reports `ANGLE_SENSOR` and refuses to move. Plug back in, `clear`.

Pass: all cycles complete with no fault, no clicking or skipping teeth, no part warm to the touch, the drum never stops BETWEEN positions. If it fails: a STALL fault means something binds: find it by turning by hand with the battery unplugged. Skipped teeth: check the mesh spacing and the frame screws. Drum stops BETWEEN: re-run the drum calibration; check the drum contacts close cleanly.

Stage 4. Springs on, no dart

Fit both springs. Start at the lowest pitch, rail pointing at a clear area.

1. `aim` to the lowest pitch. `arm` and `fire` once. Look at every part of the launch train after the shot.
2. Five more shots. Check the stop bumpers, the carriage, the rack teeth and the big gear's last tooth after each.
3. Watch `amps` while cocking: note the peak.

Pass: the carriage stops on the bumpers, nothing cracks or loosens over 6 shots, cocking current stays under the stall value, the rail holds its pitch while cocking. If it fails: cracks at the bumpers: stop and report (no more dry shots). Rail drops while cocking: the aim worm is slipping, check its mesh. STALL while cocking: the springs are stronger than planned (redo B2) or something binds.

Stage 5. Dummy dart, no tether

Make the dummy: the plate A hull top and bottom, nose and tail cap (spare prints or the dart shells before the electronics go in), taped shut with M8 nuts or sand inside to 280 g. Paint the nose bright.

1. `aim` 20 degrees. Load the dummy, check `SW_SEATED` is closed. Fire.
2. Film it side-on with the phone in slow motion, tape measure on the ground in view (muzzle speed, G5).
3. Measure the range. Repeat at 30 and 45 degrees (G6).

Pass: clean exit (no rubbing marks on the dummy), it flies nose first, range within 30 percent of the ballistics table at the same pitch. If it fails: tumbles: check the nose ballast is in the dummy too (it must match the real dart). Short range: measure the spring (B2) and the muzzle speed, then re-run the physics. Rubbing marks: check the cradle and the barrel against the dart.

Stage 6. Tether on

Tie the line through the dummy's tail cap holes. Line routed as in the base assembly steps (it bypasses the capstan for launch). Storage reel brake set: pull the line by hand with the luggage scale, 50 to 100 g (G4).

1. Fire at 30 degrees.
2. Check the line paid out with no loops or overrun on the reel.
3. `mode` REEL, `wrap`, then `reel` until `SW_SEATED` or the dummy is back near the base.
4. `unwrap`, `mode` LAUNCH. Check the line runs on the launch path again.

Pass: payout is clean, the range drop against stage 5 is under 20 percent, the wrap lays on the capstan, reel-in works, unwrap returns the line to the launch path. Repeat 5 times. If it fails: backlash loops on the reel: raise the payout drag a little (do not exceed 1 N). Big range loss: the drag is too high or the line catches; watch the slow-motion video. Wrap misses the capstan: check the wrap arm position at the end of `wrap`.

Stage 7. The anchor in soil

The real dart, built and tested to the dart run sheet section 10. Use real soil (moist loam is the target).

1. Place the dart on the soil by hand, tether tied, line slack. Send `/auto` from the operator page (or key a).
2. When the dart reports ANCHORED: pull the tether by hand with the luggage scale, slowly, up to 50 N or until it moves (G7).
3. Re-anchor. Base in REEL with wraps on: `reel` with `until` tension at a limit 10 N under the hand result.
4. Send `/release`: drills out, foot out, kick back, deploy up. Then reel the dart home.

Pass: the dart anchors on its own, holds at least 20 N in loam, the base stops at its tension limit, the release sequence frees the dart, it reels home. If it fails: augers back out: swap that motor's leads (dart run sheet section 10). Pulls out at low force: try firmer soil, and report the force (it sets the reel limit). Base overshoots the limit: redo the current calibration (G1).

Stage 8. Full mission

From the operator page on the Jetson (jetson/README.md): preflight, pick a target on the Kiyo view, aim, confirm the fire prompt, flight video from the dart, landing verdict, anchor, tow, release, reel home.

1. First run: short range (5 to 8 m), soft soil, open ground.
2. Watch every prompt: the software stops and asks before each dangerous step.

Pass: the whole sequence runs with only the expected operator confirmations, the dart lands within 2 m of the target, anchors, the rover is pulled toward the dart, the dart releases and is reeled home. If it fails: note the mission step and the fault code in the log (`mission.jsonl` on the operator page), then repeat the stage where that step was first tested.

13. Troubleshooting

First move for any problem: send `estop` (or pull the battery plug), then read the `fault` field in telemetry.

Base

Symptom	Likely cause	Fix
No reply from the base on the Jetson	Charge-only USB cable; cable in the UART port; wrong serial device	Data cable in the port marked USB. On the Jetson, `ls /dev/ttyACM*` must show a device (jetson/README.md)
ESP32 resets when the wiper starts	Ground or USB power dip	Check the ESP32 GND wire to the BTS7960 GND; short USB cable; report BROWNOUT faults
Wiper does nothing, no fault	R_EN / L_EN not wired, 20 A fuse blown	Battery unplugged, check the fuse and the two enable wires
Wiper turns one way only	RPWM or LPWM loose	Re-seat the Dupont jumper
20 A fuse blows	Stall into a hard stop, short on the wiper leads	Find the bind by turning by hand; never fit a bigger fuse
Fault STALL or OVERCURRENT	Something binds, springs stronger than planned	Battery unplugged, turn the train by hand, find the tight spot; redo the spring measurement
Fault DRUM_TIMEOUT	Drum N20 not turning: no 9 V, SLP not high, gear slipping	Meter on DRV8833 VM (9.0 V); check the SLP wire; check the N20 gear set screw
Fault DRUM_SENSOR or SWITCH_DISAGREE	A drum contact does not close or stays closed	Watch `sw` while turning the drum by the hand lever (battery unplugged for the lever, then plug in to read); bend or re-screw the contact
Fault ANGLE_SENSOR	AS5600 cable out, magnet too far	Re-seat the STEMMA cable; magnet 1 to 3 mm from the chip, centred
`pitch_deg` wrong after a restart	Pitch reference lost	Run the pitch part of the calibration again (section 10)
Drum stuck between positions with the power off	Expected: the worm holds it	Use the hand lever on the drum shaft end, battery unplugged
Base cocked and you need to unload without firing		Clear the area and fire it at the lowest pitch with no dart. If the base cannot fire, battery unplugged, and follow the unload step in the base assembly steps; never hand-release the carriage

Rover and video

Symptom	Likely cause	Fix
Jetson turns off under load	Battery low, or voltage dip	Charge the LiPo; check the barrel plug is tight; check `vbat`
No dart video	VTX and RC832 on different channels; RC832 not powered	Set the RC832 to the VTX channel (scan button); check its LED and the RCA lead
Dart video is snow	Range, antenna off, bad channel	Antenna on both ends; try another channel
Kiyo X shows nothing or errors	JPEG library mismatch on JetPack	The software forces YUYV or MJPEG (jetson/README.md); unplug and replug

Dart

Symptom	Likely cause	Fix
Serial Monitor blank	USB CDC On Boot disabled	Tools > USB CDC On Boot: Enabled, re-upload
No KEDGE WiFi	XIAO not powered or crashed	Loop plug in? 5 V at the XIAO? Press reset
A motor turns the wrong way	Leads reversed	Loop plug out, swap that motor's pair (run sheet section 10)
Current trip during anchoring	Stall on a stone, or the trip set too low	Re-measure stall currents (section 14 F3) and update the tuning
INA219 or BNO085 not found	I2C wiring	`z` key scans; SDA and SCL must not be swapped

14. Measure list

From `docs/build/MEASURE_LIST.md`*:*

What to measure or confirm

Everything nobody could know from here. Do group A and B first: the base parts that depend on them should not be printed until those numbers are in `base/spec/params.py` and the pipeline has been re-run (see "After you measure" at the end). Tools: caliper, multimeter, kitchen scale, luggage scale, phone camera, tape measure.

Photo rule: for every caliper reading, take a photo with the caliper on the part and the reading visible. Put the photos in `originals/measure_2026-11/` so the numbers can be checked later.

A. Wiper motor (params.py `WIPER`, set `measured=True` when done)

#	What	How	Changes	Reprint after
A1	Output shaft diameter	Caliper across the round part of the shaft, 3 places, take the smallest	`WIPER.shaft_d`	Wiper shaft coupler (drum parts)
A2	Shaft flats or spline	Flats: caliper across the flats (`shaft_flat`). Spline: count the teeth, caliper the outside and root diameter, photo end-on	`WIPER.shaft_flat` (note a spline in REPORT)	Wiper shaft coupler
A3	Shaft end thread and length	Thread a nut you know (M6, M8) onto it. Caliper the shaft length from the housing face to the tip	`WIPER.shaft_thread`, `WIPER.shaft_len`	Wiper shaft coupler
A4	Mount hole pattern	Lay the motor face down on paper, press so the holes mark it (or trace with a pencil). Then caliper each hole centre from the shaft centre along two square directions and each hole diameter. Photo of the paper with a ruler	`WIPER.mount_holes` as (x, y, dia) from the shaft axis	Motor adapter plate (frame parts)
A5	Housing size	Caliper the gearbox box (length, width, height) and the motor can (diameter, length), and how far the shaft sits from each edge	`WIPER.body_box`	Frame cheeks and guard if they clash (the audit says)
A6	Free speed and free current at 12 V	Motor clamped, nothing on the shaft, LiPo through lead #2 (20 A fuse). Meter on the 10 A range in series with the red wire. Tape flag on the shaft: count turns for 30 s, rpm = turns x 2. Note the amps	`WIPER.n0_rpm`, `WIPER.free_a`	Nothing; re-run physics
A7	Stall current and stall torque	Do NOT use the meter: stall is about 18 A, over the meter's 10 A range. After base calibration (firmware/base/CALIBRATION.md), use the `amps` telemetry. Torque: clamp a 300 mm bar on the shaft, hook the luggage scale at the end, jog the motor for at most 2 s at full duty and read the peak. Torque N m = kg x 9.81 x 0.30	`WIPER.stall_a`, `WIPER.stall_nm`	Nothing; re-run physics and the ballistics table

B. Launch springs (params.py `SPRING`, set `measured=True` when done)

The two springs are extension springs with initial tension: they need a small force before they stretch at all, so one weight is not enough. Use two loads.

#	What	How	Changes	Reprint after
B1	Free length L0	Caliper the spring hook to hook (inside of the hooks), no load	`SPRING.free_len`	Nothing
B2	Rate k and initial tension F0	Weigh a full 1 L bottle (W1, kg) and a full 2 L bottle or two 1 L bottles in a bag (W2). Hang W1 from one spring, measure L1 hook to hook; then W2, measure L2. k (N/mm) = (W2 - W1) x 9.81 / (L2 - L1). F0 (N) = W1 x 9.81 - k x (L1 - L0). If L1 equals L0 (the spring did not move), use W2 and a 4 L load instead. Do both springs	`SPRING.k_n_per_mm`, `SPRING.F0_n`	Nothing unless the stroke changes (physics says); re-run physics and the ballistics table
B3	Wire and outside diameter	Caliper	`SPRING.wire`, `SPRING.od`	Spring anchors only if OD differs by more than 0.5 mm

C. Parts box: confirm you have these (no new purchases)

#	Item	Needed for	If you do not have it
C1	608 bearings (8 to 10)	storage reel, capstan or wrap arm, cam roller	The only purchase: search "608ZZ bearing 8x22x7"
C2	M8 bolts or 8 mm steel rod, and their lengths	axles through the 608s	Printed PPA-CF axles (tell the coordinator, it is a params switch)
C3	Worm N20 count: you need 4 (dart kick, deploy, foot, and the base selector drum)	dart and base	The kedge notes said 2 in hand and one to buy; the August order listed 8 worm N20s. Count them
C4	MP1584 #2 (the spare from the dart build)	drum N20 supply at 9.0 V	No other way to run the DRV8833 from the 3S: tell the coordinator
C5	DRV8833 #3 (freed from the dart by the new pin map)	drum N20	
C6	100 kg load cell + HX711 (August order)	optional line tension reading	Leave it out: the firmware uses wiper current instead
C7	Resistors: 3 x 10 k, 1 x 1 k, 1 x 680 R, 1 x 100 k, 1 x 22 k	BTS7960 enable pull-downs, DRV8833 SLP pull-down, IS divider, battery divider (firmware/base/WIRING.md)	Any value within 20 percent works for the pull-downs; tell fw-base if the dividers differ
C8	Spare XT60 plugs and a fuse holder, 14 AWG wire	XT60 Y harness	
C9	Barrel plug 5.5 x 2.5 mm with lead	Jetson from the LiPo	Run the Jetson from its own supply on the bench; the rover needs one
C10	Yellow RCA lead	RC832 AV out to the DIGITNOW	
C11	USB data cable for the DevKitC, STEMMA QT to Dupont cable	base ESP32, AS5600	
C12	M3 screws (lengths from the base print list), M3 nuts, M3 heat-set inserts	printed contacts, frame joints	

D. Electrical labels to read

#	What	How	Changes
D1	RC832 input voltage	Read the label or the listing. Most take 7 to 24 V. If it says 12 V only, it still runs from the 3S (12.6 V full) but report it	SYSTEM_ELECTRICAL.md run P8
D2	Jetson DC input range	Label on the Jetson's own supply and the NVIDIA dev kit page. The 3S gives 9.9 to 12.6 V in use	SYSTEM_ELECTRICAL.md run P7
D3	DRV8833 carrier pad names	Which pad is SLP / nSLEEP / EEP on your boards. Meter on beep: that pad must NOT beep to VM / VIN	firmware/base/WIRING.md
D4	DevKitC board version	Printed on the back (v1.0 or v1.1). The RGB LED pin differs; the firmware avoids both	Nothing
D5	Video standard	When the dart video shows on the Jetson, note the size (720 x 480 = NTSC, 720 x 576 = PAL)	jetson config

E. Rover

#	What	How	Changes	Reprint after
E1	Deck size and mounting holes	Tape measure the free deck area; caliper the hole positions and diameters from one corner	params.py `FRAMEWORK` (rover mount)	Base deck / rover mount
E2	Drive train and whether the Jetson drives it	Tell the coordinator what drives the wheels and what controls it	jetson rover module	Nothing
E3	Kiyo X height and tilt on the rover	Tape measure lens to ground; phone level app on the camera for tilt	jetson camera model (target calibration)	Nothing
E4	Base launch height	Tape measure from the ground to the main shaft axis when the base is mounted	physics (ballistics table)	Nothing

F. Dart (from the KEDGE notes, still open)

#	What	How	Changes
F1	Dart v30 is the one being built	Confirm	print pack choice
F2	Bought part weights and the finished dart weight	Kitchen scale	`DART.mass_kg`, dart physics

#	What	How	Changes
F3	Stall current of each dart motor at 2S	The dart `i` key while holding the shaft with pliers for 1 s	firmware/dart tuning values (see firmware/dart/README.md)
F4	OTOTEC BMS size and F693ZZ flange	Caliper (pocket is 40.8 x 8 x 2.1 plus 0.5 each way)	dart geometry if it does not fit
F5	Nose ballast	Weigh it: 15 g	flight stability

G. Bench and field numbers (measured during the TEST_PLAN stages)

#	What	How	Changes
G1	BTS7960 current sense scale	firmware/base/CALIBRATION.md (meter in series at low duty, compare with `amps`)	firmware `set` key (stored on the ESP32)
G2	Battery voltage reading	Meter on the LiPo vs `vbat` telemetry	firmware `set` key
G3	AS5600 zero and drum positions	firmware/base/CALIBRATION.md	stored on the ESP32
G4	Storage reel payout drag	Pull the line off the reel slowly with the luggage scale in LAUNCH mode: 0.5 to 1 N (50 to 100 g). Above 1 N costs range, 3 N ruins the flight	slip clutch preload (reel parts)
G5	Muzzle speed	Phone slow-motion video side-on with a tape measure in view: speed = distance moved / time between frames	physics model, ballistics table
G6	Range at three pitches	Tape measure from the base to the dart at 3 pitches, no tether brake changes	ballistics table check (jetson uses it)
G7	Soil holding force	Dart anchored, luggage scale on the tether, pull slowly up to 50 N, note where it moves	reel tension limit

After you measure

1. Write the numbers into `base/spec/params.py` (set `measured=True` in `WIPER` and `SPRING`) or send them to the coordinator. Put photos in `originals/measure_2026-11/`.
2. Re-run the base pipeline (assembly, audit must say CLEAN, physics, ballistics table, print pack) and `firmware/base/tools/gen_params.py`, then re-flash the base ESP32.
3. Reprint only the parts named in the "Reprint after" column. If you have not printed them yet, print after step 2.

Numbers still assumed in params.py right now

params.py version: `0.1-stub`.

params.py name	State	Who fixes it
WIPER	assumed, not measured yet	MEASURE_LIST
SPRING	assumed, not measured yet	MEASURE_LIST
LAYOUT	not decided yet (None)	coordinator
DRUM	not decided yet (None)	coordinator
LAUNCH	not decided yet (None)	coordinator
AIM	not decided yet (None)	coordinator
REEL	not decided yet (None)	coordinator
FRAMEWORK	not decided yet (None)	coordinator
SWITCHES	not decided yet (None)	coordinator
TIMELINE	not decided yet (None)	coordinator

15. Sources and appendices

This guide is built by `docs/build/make_guide.py` from the files below. A "NOT YET" row is a section that still shows a placeholder; re-run the script after it lands.

Source	Written by	In the repo
`base/spec/BASE_SPEC.md`	coordinator	yes
`base/pack`	cad-infra (print pack)	NOT YET
`base/out`	cad-infra (assembly pipeline)	NOT YET
`base/guide`	cad-infra (assembly steps)	NOT YET
`kedge/pack_v30/KEDGE_v30_run_sheet.pdf`	dart v30 pack (done)	yes
`firmware/base/WIRING.md`	fw-base	NOT YET
`firmware/base/README.md`	fw-base	NOT YET
`firmware/base/CALIBRATION.md`	fw-base	NOT YET
`firmware/base/BENCH_TEST.md`	fw-base	NOT YET
`firmware/base/PROTOCOL.md`	fw-base	NOT YET
`firmware/dart/WIRING.md`	fw-dart	NOT YET
`firmware/dart/README.md`	fw-dart	NOT YET
`firmware/dart/API.md`	fw-dart	NOT YET
`firmware/dart/BENCH_TEST.md`	fw-dart	NOT YET
`jetson/README.md`	jetson	NOT YET

Appendices attached after this page (when present): the base assembly steps PDF from `base/guide`, then the KEDGE v30 dart run sheet. In the run sheet, follow section 6 of this guide for the new dart pin map.

Other documents in the repo: `kedge/pack_v30/KEDGE_v30_engineering_report.pdf` (dart physics and loads), `base/spec/BASE_SPEC.md` (base design), `firmware/base/PROTOCOL.md` (base serial protocol), `firmware/dart/API.md` (dart HTTP API), `jetson/INTERFACES.md` (Jetson software).

KEDGE dart v30: build day run sheet

Dart: Ø42 x 152 mm, 1.5 mm walls, PLA on the X2D. Kick leg 0 to 201.75°, drill pair 0 to 42.25°. Collision sweep clean (kick in 102 steps to 201.75°, drills in 1° steps to 42.25°, every sub-part, exact meshes). Printed volume 91.91 cm³ (about 114.0 g if solid; less at 40% infill). About 12 hours wall clock with the two plates printing in the background. Rules for the whole day: the power loop plug stays OUT while you solder, move a wire or plug in USB. Never let the two cells touch metal together. One step at a time; each step says what you should see before you move on.

1. What is in the pack

File	What it is
print/plate_A_hull.3mf	Print plate A: hull top, hull bottom, nose, tail cap, both couplers. Already in print orientation.
print/plate_B_legs_tray.3mf	Print plate B: tray, kick leg, both drill pods, three augers. Already in print orientation.
print/fit_coupon.stl	20 minute test print: checks every fit before the long plates (section 4).
print/*.stl, step/*.step	Every printed part on its own, oriented for printing. STEP is the editable CAD (augers are faceted in STEP).
docs/wire_cut_list.csv	All 53 wires, pad to pad, colour, gauge and cut length (also section 9 of this sheet).
firmware/kedge_dart/	Firmware v3: kedge_dart.ino + kedge_logic.h. Keyboard over USB and WiFi, telemetry /status for the rover, current trip, auto-anchor (foot spins while the kick plants it, drills spin while the pods deploy), auto-on-landing. Compiled on ESP32 core 2.0.17 and 3.3.12.
firmware/test/	PC simulation of the control logic (all tests pass).
firmware/reel/reel_tension_limit	Reel sketch for the DevKitC + BTS7960: reel-in speed tapers and stops at a set tether tension (motor current or the load cell).
firmware/jetson/kedge_client.py	Rover / laptop client: status, logging, hold-a-key, auto, terminal driving, FPV overlay.
docs/KEDGE_*_engineering_report.pdf	Flight, launch and landing loads, holding force by soil, reel limit, launcher silhouette, design versions.
launcher/barrel_profile.dxf / barrel_keepout.stl	What the launcher barrel has to clear, with 1 mm clearance.
source/	The parametric generator and the whole check pipeline (run_all.sh). Every number lives in P at the top; KOV overrides make variants.

2. Parts and tools to have on the bench

Item	Qty	Note
XIAO ESP32S3, DRV8833 x3, INA219, BNO085, MP1584 (the 5.04 V one)	1 each	Headers clipped off (flush cutters), INA219 screw terminal cut off. MP1584 #2 stays spare.
OTOTEC 2S BMS stick	1	Caliper it when it arrives: the pocket under the cells is 40.8 x 8 x 2.1 plus 0.5 each way.
18350 cells, charged	2	Charge both to 4.2 V in the XTAR first.
Worm N20 (kick, deploy) + third worm N20 (foot, buy)	3	Buy the foot one single-shaft if listed; otherwise cut its lower stub to 2 mm.
Drill motors (12 x 10 x 24.27, M3 x 56 shaft)	2	
F693ZZ flanged bearings	5	2 per pod + 1 in the kick ear (bag has 10).
M3 nuts / M3 nylocs	2 / 2	Nut against each pod's inner bearing, nyloc in each auger.
M3 x 8 screws	7 + 2	7 countersunk (flat head, DIN 7991) for the radial hull screws, they sit flush with the skin; 2 pan head for the pod retainers (with washers).
M3 x 12 screw, M3 x 10 screw	1 + 1	x12 goes through the ear root into tray disc 1; x10 is the kick hub screw.
M3 washers	3	2 pod retainers + 1 between kick hub and ear bearing.
M3 grub screw	1	Foot auger on the foot N20 shaft flat.
2xAA battery holders (for their contacts)	2	You need 2 springs and 2 flat plates. Launch throws the cells backwards, so springs go at the FRONT of the cell bay and flat plates at the BACK.
470 to 1000 µF capacitor, 10 V or more	1	Across the MP1584 5 V output: rides through contact bounce on landing.
Schottky diode 1N5817 or SS14	1	In wire 18 (5 V to the XIAO): stops back-feeding USB.
Nose ballast 15 g		Steel shot or 3 to 4 M6 nuts set in 5-minute epoxy in the nose cavity. Without it the dart tumbles unless the tether is taut (engineering report, section 3).
JST-XH 3-pin pair, JST-XH 2-pin pair + a 2-pin shorting plug	1 each	Charge socket and power loop in the tail window.

Item	Qty	Note
Silicone wire 22, 24, 26, 28 AWG		Totals per colour in section 9.
Zip ties 2.5 mm, medium threadlocker (or CA), double-sided foam tape, hot glue		
Already on your bench		Caddx Ant camera, Zeus VTX + u.FL antenna, 3S LiPo + XT60 lead #1, XTAR MC2 (charges the 18350s out of the dart), breadboard and jumpers.
Tools		Soldering iron, solder, flux, multimeter, wire strippers, flush cutters, small flat screwdriver (MP1584 trimmer), heat shrink + lighter or heat gun, JST-XH crimper or pre-crimped pigtails, 5-minute epoxy, hex keys, tweezers, luggage scale (pull test).

3. Timeline

Clock	Step	Printer
0:00	4. Fit coupon	coupon (20 min)
0:30	5. Start plate A, flash the firmware	plate A (about 3 h)
1:00	6. Prepare boards, set both MP1584s, cut and label all wires	plate A
2:00	7. Solder the bench harness, power-up test, motor directions	plate A
3:30	Start plate B. Clean plate A parts, test-fit the hull, nose ballast (8a)	plate B (about 2.5 h)
4:00	8b. Battery contacts and BMS wiring off the dart	plate B
6:00	8c. Pods, couplers, augers (plate B done)	
7:00	8d. Populate the tray, battery power-up check	
8:30	8e. Close the hull, couplers, pods	
9:30	8f. Kick leg and foot	
10:15	10. Function test and calibration	
12:00	Done. Charge the cells (section 10, Charging).	

Bambu Studio settings (both plates): PLA, 0.4 nozzle, 0.16 mm layers, 4 wall loops (the 1.5 mm walls come out solid), 40% gyroid infill, top/bottom 5 layers, supports OFF, brim ON (auto) for the augers and couplers, X-Y hole compensation from the coupon (section 4). Plate times above are estimates; Studio shows the real ones.

Part	Print pose	Size on bed mm	Volume cm³
hull_top	split face down (on the bed), saddle up	121 x 50 x 35	17.41
hull_bottom	split face down, belly up	121 x 42 x 21	11.5
nose	tip down on the bed (flat Ø22), spigot up	42 x 42 x 27	8.25
tail_cap	end face down, spigot up (no supports)	42 x 42 x 15	7.13
tray	keel down on the bed; discs stand up, decks bridge 26 mm between discs	109 x 39 x 36	15.98
kick_leg	on its side (pole axis horizontal, hub face down)	119 x 15 x 15	8.7
drill_leg_L	upright, motor pocket open to the top (as installed folded)	46 x 15 x 13	3.37
drill_leg_R	upright, motor pocket open to the top (as installed folded)	46 x 15 x 13	3.37
drill_leg_L_auger	base down, nut slot sideways	17 x 17 x 73	5.5
drill_leg_R_auger	base down, nut slot sideways	17 x 17 x 73	5.5
foot_auger	base down	17 x 17 x 40	3.06
coupler_L	standing on its socket end	7 x 6 x 36	1.24
coupler_R	standing on its socket end	7 x 6 x 27	0.9

4. Fit coupon (0:00, 20 minutes)

Print fit_coupon.stl. It has, left to right: the worm N20 pocket, the drill motor pocket, the F693ZZ seat with its flange recess, the hull boss bore, the N20 D-bore, an M3 pilot, an M3 clearance hole, the pod D-bore, the 3.8 shaft clearance, a coupler peg, and a hull ring with its spigot ring.

Check	Should feel	If not
Worm N20 into the square pocket	Slides in by hand, no wobble	Tight: hole compensation +0.05. Loose: -0.05.
Drill motor into its pocket	Slides in by hand	Same as above
F693ZZ into the seat	Firm thumb press, flange flush in the recess	Too tight: +0.05. Falls out: a drop of CA on assembly day.
Coupler peg into the boss bore	Turns freely, little play	Tight: sand the peg lightly
Coupler peg D-socket onto an N20 shaft	Pushes on firmly, no rotation slop	Loose: a drop of CA on assembly
Coupler peg into the pod D-bore	Slides in, no rotation slop	
M3 screw into the 2.5 pilot	Cuts its own thread, holds	
Spigot ring into the hull ring	Push fit, no rattle	Loose: one layer of tape on the spigots

Directions used below: left = the kick-leg side, seen from the tail with the saddle on top (y plus in the CAD). The left pod is the brighter red one in the viewer.

5. Start plate A and flash the firmware (0:30)

1. Send plate_A_hull.3mf to the X2D.
2. In the Arduino IDE: board XIAO_ESP32S3, Tools > USB CDC On Boot: Enabled (the sketch will not compile otherwise). Library Manager: install Adafruit INA219 and Adafruit BNO08x.

3. Nothing else connected to the XIAO. Plug it in by USB-C, open firmware/kedge_dart/kedge_dart.ino, Upload. Re-select the board if the IDE switches it to the wrong one.
4. Serial Monitor at 115200, 'No line ending'. You should see the KEDGE ready help line (INA219 and BNO085 'not found' is normal now). If the monitor stays blank: press the XIAO reset button once, check the port and USB CDC On Boot.
5. On your phone or laptop, join WiFi KEDGE (password grappstar) and open 192.168.4.1: you get the control page. From now on you can also upload over WiFi: Arduino IDE > Port > the network port called kedge. Unplug USB.

6. Prepare boards and wires (1:00)

1. Clip every header pin flush on the XIAO and the three DRV8833 boards (flush cutters, close to the board). Cut the INA219 screw terminal off. Tin each pad you will use.
2. MP1584s, with the 3S LiPo on XT60 lead #1 and the multimeter on the output, turning the small brass screw with a small screwdriver: the dart one to 5.04 V; the spare one to 8.0 V (this is your bench motor supply in step 7). Disconnect.
3. Cut all wires to the lengths in section 9. Strip 3 mm each end, tin, and put a tape flag with the wire number on each. Always slide heat shrink onto a wire before you solder its second end.
4. DRV8833 check before any 3V3 wire: the driver chip has no logic supply pin. The pin called VIN (or VCC on some red boards) is the MOTOR supply. The 3V3 wires (25 to 27) go only to the SLP / nSLEEP / EEP pad, which wakes the driver. Meter on beep: touch the pad you plan to use and the VIN/VM pin. If it beeps, it is the motor supply: do not put 3V3 on it, or the battery voltage reaches the XIAO.
5. Diode: wire 18 has a Schottky diode (1N5817 or SS14) in it, stripe toward the XIAO, in heat shrink. It stops the 5 V buck back-feeding your Mac when USB is plugged in. Still never plug USB in while the dart is powered.

7. Solder the bench harness and test it (2:00)

Lay the boards on the bench in the order they sit in the dart (picture in section 11): sled boards (MP1584, DRV#3), XIAO + DRV#1, DRV#2 + INA219, BNO085. Solder in this order and check after each group:

1. Power: wires 10 to 13 and 15 to 24, and wire 14 at its MP1584 end only (its BMS end comes in 8b). Fit the capacitor across MP1584 OUT+ / OUT-: long leg to OUT+, the striped leg to OUT-. Snap the u.FL antenna onto the VTX now: never power a VTX without its antenna. Also solder wire 9 at its INA219 VIN+ end only (its socket end is crimped in 8b). Meter on beep: red rail to black rail must NOT beep.
2. 3V3 and ground to the logic: wires 25 to 31.
3. I2C: wires 32 to 35. Motor control: wires 36 to 43. Bridge AIN1 to BIN1 and AIN2 to BIN2 on DRV#2 with a short wire offcut.
4. Motor leads: solder wires 44 to 53 to the DRV outputs only; leave the far ends free.
5. Power-up test with the bench supply (the 8.0 V MP1584 on the LiPo): its OUT+ to the INA219 VIN+ pad, OUT- to the MP1584 IN- pad (main ground). You should see about 4.7 V on the XIAO 5V pad (5.04 minus the diode), the VTX LED on, and on the WiFi page: i reads about 8 V; z finds 0x40 and 0x4a. Disconnect the bench supply.
6. Motor directions, bench supply on again. Twist each motor lead temporarily onto its wire (no solder yet: the kick, drill and foot leads are threaded through holes later). Hold each key half a second and check:

Key	Motor	Must turn so that	If not
1	deploy N20	the output shaft turns the way that swings the pods DOWN (mark the shaft with a pen and compare with the viewer)	swap the pair on wire 44 / 45
2	kick N20	the hub would swing the leg UP over the nose	swap 46 / 47
3	drills	each drill output shaft turns: left motor clockwise seen from behind the motor, right motor anticlockwise (the augers are mirror images, so both then screw in)	one wrong: swap that motor's pair; both wrong: swap 40 / 41
4	foot N20	the foot N20 output shaft turns clockwise, seen from the shaft end (that screws the right-hand foot auger in)	swap 52 / 53

Note the free current on i for each (October bench: worms 20 to 30 mA, drill pair 60 to 70 mA). Mark each motor's lead that goes to the lower wire number. Solder the deploy N20 now (it never goes through a hole). Disconnect the bench supply.

8. Mechanical assembly

8a. Nose ballast (3:30, plate A done)

Stand the nose on its tip. Drop 15 g of steel shot or 3 to 4 M6 nuts into the cavity around the camera tube (not into the camera pocket) and cover them with 5-minute epoxy. Leave it to set while you carry on.

8b. Battery contacts and BMS, off the dart (4:00)

1. Join the two springs with wire 2 (the series link, 22 AWG) and solder wire 3 (BMS BM) to that link. Solder wire 1 to one flat plate and wire 4 to the other.
2. On the bench, solder wires 1, 3 and 4 to the BMS B-, BM and B+ pads, wire 14 to P-, wire 8 to P+, and wires 5 to 7 to B-, BM, B+ for the charge socket. Crimp or solder the JST-XH sockets on wires 5 to 7 (3-pin) and 8 / 9 (2-pin power loop).

8c. Drill pods, couplers and augers (6:00, plate B done)

1. Press one F693ZZ into the inner seat from the motor pocket side, flange toward the motor. Press the other into the nose end from outside, flange on the nose face.
2. Thread an M3 nut onto the drill motor shaft, about 3 mm from the motor face. Feed the shaft through both bearings, drop the motor into its pocket. Threadlocker on the shaft, then hold the nut with tweezers and turn the motor until the nut seats against the inner bearing. Zip-tie the motor in its groove.
3. Push an M3 nyloc into the side slot of the auger (nylon ring toward the point). Augers are handed: the LEFT pod takes drill_leg_L_auger (right-hand flight), the RIGHT pod takes drill_leg_R_auger (left-hand flight). Threadlocker on the shaft end, screw the auger on until its base boss touches the outer bearing.
4. Check: each auger spins freely with the motor, no rub on the pod nose.

8d. Populate the tray (7:00)

1. Glue the joined springs into the long recess on the FRONT cell-bay disc (nose side), one spring in front of each cell position. Glue the wire 1 plate into the BACK disc recess on the right (cell 1 side) and the wire 4 plate on the left. Lay the BMS in the channel under the cell bay.
2. Deploy N20 into the sled cradle (worm box toward the tail, output shaft across), zip tie through the groove under the cradle.
3. Stick the boards on with foam tape: DRV#3 on the keel and MP1584 on the shelf in the sled; BNO085 on deck A; XIAO (USB toward the nose) and DRV#1 on deck B; DRV#2 and INA219 on deck C; VTX on the deck over the cells. Route bundles through the notches in the discs. Antenna lead to the tail.

4. Push both sockets into the tail cap window, antenna through the top slot (wire 9 and wire 14 are already soldered from 7.1 and 8b). Tape over the charge socket now: its pins are live as soon as the cells go in.
5. Cells in: cell 1 (right) button toward the nose, cell 2 (left) button toward the tail. Each pushes in sideways, squeezing its spring, and sits hard against its back plate.
5a. Loop still out. Meter on the charge socket pins (lift the tape corner): pin 1 to 2 reads 3.6 to 4.2 V, pin 2 to 3 reads 3.6 to 4.2 V, pin 1 to 3 reads 7.2 to 8.4 V. Any negative reading: pull the cells out and check their direction.
6. Battery power-up check: check the u.FL antenna is still clicked onto the VTX, then plug the loop in. WiFi page: i reads 7.4 to 8.4 V; z finds 0x40 and 0x4a; the deploy N20 runs on 1. Pull the loop plug.

8e. Close the hull (8:30)

1. Kick N20 into the saddle on the hull top, worm box toward the nose, output shaft through both side holes. Zip tie: down one slot beside the saddle, under the skin, up the other, over the motor. Its leads go down through the hole in the saddle floor.
2. Camera into the nose pocket from the back (lens into the hole), cable out the back. Tray into the bottom half, and seat the nose spigot in the bottom half at the same time: the groove on the spigot sits over the half-rib inside the hull.
3. Thread the leads before the top half goes on: kick leads (wires 46, 47) up through the saddle floor hole; foot leads (52, 53) out through the skin hole beside the saddle; drill leads (48 to 51) out through the two flank holes. Lower the top half on.
4. Radial screws into disc 1: M3 x 8 at left z -3, right z +3 and right z -3, and the M3 x 12 through the ear root at left z +3; four M3 x 8 into the tail spigot (z ±3 both sides). The top half's rib closes the nose groove, so the nose is locked.
5. Solder the kick N20 and the two drill motors to their leads (as marked in 7.6), heat shrink (slide it on before soldering; iron on each joint 3 s at most; shrink with a heat gun on low, no lighter near the PLA).
6. Couplers: slide coupler_L in through the left boss, coupler_R through the right, turning gently until each D-socket drops onto the deploy N20 stub. Pods onto the coupler ends (D-bore), then M3 x 8 + washer into each coupler end. Fold the pods along the flanks.

8f. Kick leg and foot (9:30)

1. Press an F693ZZ into the ear from outside, flange in the recess.
2. Foot N20 into the foot housing (opening on the inboard side), output shaft up through the top hole, zip tie. Foot auger onto the shaft, grub screw onto the flat.
3. Feed the foot leads from the foot pocket through the pole channel and out the small hole near the hub. Solder them to wires 52 and 53 (as marked in 7.6): heat shrink on first, iron 3 s at most per joint, heat gun on low, no lighter near the PLA. Leave the loop round the hub.
4. Push the hub onto the kick N20 stub (D flat), put the washer between hub and ear, M3 x 10 through the ear bearing into the hub. Plug the loop in and send @ until the leg rests folded on the hull.

10. Function test and calibration (10:15)

Use a tray of damp soil or sand. Laptop on the KEDGE WiFi page; hold a key to run, let go to stop. Space bar stops everything.

Key	Should happen	If it goes the wrong way
First, in the air	Hold each key for half a second: directions were fixed in 7.6	If one is still wrong, swap it at the DRV end (open the hull)
2 (hold)	Kick leg swings up over the nose and down; the foot auger tip meets the soil near 180°	Swap wires 46 and 47
2 and 4 (hold both)	Foot auger screws IN while the kick keeps pushing, until the foot housing sits on the soil	Foot backs out: swap wires 52 and 53
1 and 3 (hold both)	Both pods swing down and back while BOTH augers spin and cut in; the dart does not twist	Pods go up: swap 44 and 45. One auger backs out: swap that motor's pair (48/49 or 50/51). Both: swap 40 and 41 at the XIAO
i during each	Note the free current, the current when the foot touches, and the stall current	Put them in kedge_logic.h (kickContactMa, kickStallMa, deployStallMa) and upload over WiFi
b	Roll and pitch of the planted dart	
a	Full auto sequence, same order as above, by itself	Space aborts
A then throw	Arms auto-anchor on landing (the dart anchors 1.5 s after it comes to rest)	Press A again to disarm

Charging: either take the cells out and use the XTAR MC2, or plug a 2S balance charger (JST-XH 3-pin) into the charge socket in the tail. Loop plug OUT while charging.
Recovery: # (drills out), ! (pods up), $ (foot out), @ (kick back). The worms hold every position with the power off.
Note: STOP (space, /stop, or a client hold ending) also aborts a running auto sequence; that is deliberate.
Pin change from the October bench: the drill pair is now on D8/D9 and the foot on D6/D7. D6/D7 are the XIAO's serial pins and flicker at every reset; the slow foot worm does not care, the drills would twitch.

9. Wire cut list (pad to pad)

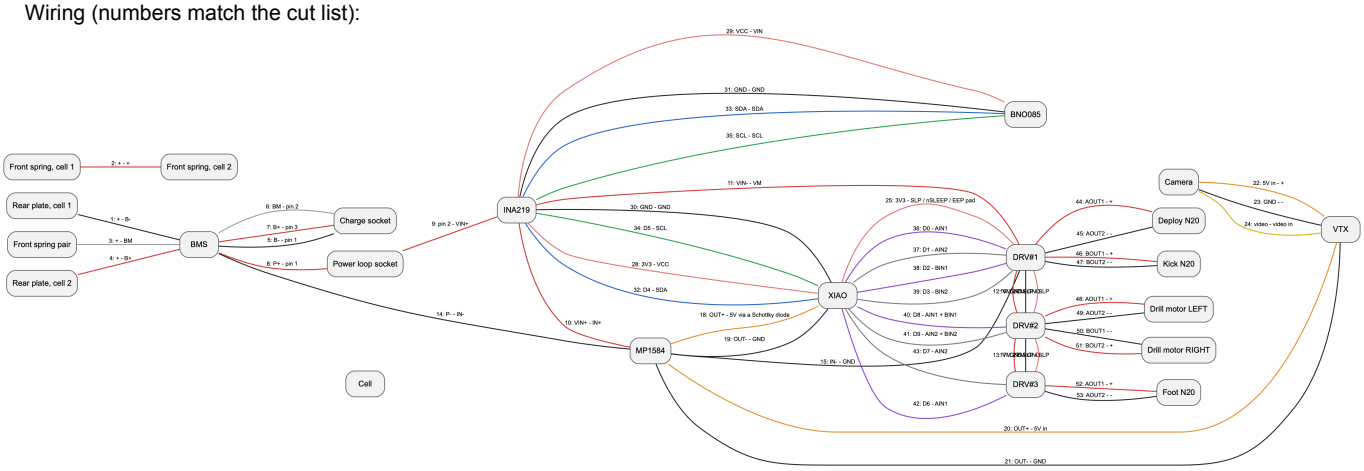
#	From	To	Colour	Gauge	Cut mm	Note
1	Rear plate, cell 1 (-)	BMS B-	black	22 AWG	45	solder to the plate tab before it goes in its recess
2	Front spring, cell 1 (+)	Front spring, cell 2 (-)	red	22 AWG	45	the series link: carries the full battery current; glue both springs in the front recess
3	Front spring pair (cell 1 + / cell 2 -)	BMS BM	white	26 AWG	55	the two springs are joined by a short wire; this is the midpoint sense
4	Rear plate, cell 2 (+)	BMS B+	red	22 AWG	45	
5	BMS B-	Charge socket pin 1 (tail window)	black	26 AWG	80	JST-XH 3-pin, balance charge
6	BMS BM	Charge socket pin 2	white	26 AWG	125	
7	BMS B+	Charge socket pin 3	red	26 AWG	80	
8	BMS P+	Power loop socket pin 1 (tail window)	red	22 AWG	80	JST-XH 2-pin; the shorting plug is the ON switch
9	Power loop socket pin 2	INA219 VIN+	red	22 AWG	95	main + after the loop
10	INA219 VIN+ (same pad)	MP1584 IN+	red	22 AWG	140	buck taps before the shunt, so INA219 reads motor current only
11	INA219 VIN-	DRV#1 VM	red	22 AWG	50	motor rail 6.0-8.4 V
12	DRV#1 VM	DRV#2 VM	red	22 AWG	65	daisy chain
13	DRV#2 VM	DRV#3 VM	red	22 AWG	155	forward into the sled (keel)
14	BMS P-	MP1584 IN-	black	22 AWG	130	main ground
15	MP1584 IN- (same pad)	DRV#1 GND	black	22 AWG	120	
16	DRV#1 GND	DRV#2 GND	black	22 AWG	65	
17	DRV#2 GND	DRV#3 GND	black	22 AWG	155	
18	MP1584 OUT+ (5.04 V)	XIAO 5V via a Schottky diode (stripe toward the XIAO)	orange	24 AWG	160	set and check 5.04 V BEFORE connecting; the diode stops the buck back-feeding USB (1N5817 or SS14, in heat shrink)
19	MP1584 OUT-	XIAO GND	black	24 AWG	160	
20	MP1584 OUT+	VTX 5V in	orange	26 AWG	95	
21	MP1584 OUT-	VTX GND	black	26 AWG	95	
22	VTX 5V in (same pad)	Camera + (red)	orange	28 AWG	105	runs forward along the top through the disc notches
23	VTX GND (same pad)	Camera - (black)	black	28 AWG	105	
24	Camera video (yellow)	VTX video in	yellow	28 AWG	105	twist with the camera ground
25	XIAO 3V3	DRV#1 SLP / nSLEEP / EEP pad (NOT VCC or VIN)	red/white	28 AWG	55	wakes the driver. Check first: if the pad you picked beeps to VIN/VM on the meter, it is the motor supply: do NOT connect 3V3 to it
26	DRV#1 SLP	DRV#2 SLP	red/white	28 AWG	65	
27	DRV#2 SLP	DRV#3 SLP	red/white	28 AWG	155	
28	XIAO 3V3	INA219 VCC	red/white	28 AWG	60	
29	INA219 VCC	BNO085 VIN	red/white	28 AWG	60	
30	XIAO GND	INA219 GND	black	28 AWG	60	
31	INA219 GND	BNO085 GND	black	28 AWG	60	
32	XIAO D4	INA219 SDA	blue	28 AWG	60	I2C SDA
33	INA219 SDA	BNO085 SDA	blue	28 AWG	60	
34	XIAO D5	INA219 SCL	green	28 AWG	60	I2C SCL
35	INA219 SCL	BNO085 SCL	green	28 AWG	60	
36	XIAO D0	DRV#1 AIN1	purple	28 AWG	55	deploy
37	XIAO D1	DRV#1 AIN2	grey	28 AWG	55	deploy
38	XIAO D2	DRV#1 BIN1	purple	28 AWG	55	kick
39	XIAO D3	DRV#1 BIN2	grey	28 AWG	55	kick
40	XIAO D8	DRV#2 AIN1 + BIN1 (bridge the two pads)	purple	28 AWG	45	drills (moved off D6/D7: those twitch at reset)
41	XIAO D9	DRV#2 AIN2 + BIN2 (bridge the two pads)	grey	28 AWG	45	drills
42	XIAO D6	DRV#3 AIN1	purple	28 AWG	145	foot (slow worm, harmless if it twitches at reset)
43	XIAO D7	DRV#3 AIN2	grey	28 AWG	145	foot

#	From	To	Colour	Gauge	Cut mm	Note
44	DRV#1 AOUT1	Deploy N20 + (red)	red	26 AWG	165	motor sits in the sled cradle
45	DRV#1 AOUT2	Deploy N20 - (black)	black	26 AWG	165	
46	DRV#1 BOUT1	Kick N20 + (red)	red	26 AWG	150	out through the hole in the saddle floor
47	DRV#1 BOUT2	Kick N20 - (black)	black	26 AWG	150	
48	DRV#2 AOUT1	Drill motor LEFT + (red)	red	26 AWG	195	out the left flank hole, 30 mm loop for the pod swing
49	DRV#2 AOUT2	Drill motor LEFT - (black)	black	26 AWG	195	
50	DRV#2 BOUT1	Drill motor RIGHT - (black) SWAPPED	black	26 AWG	170	reversed on purpose: the pair counter-rotates
51	DRV#2 BOUT2	Drill motor RIGHT + (red) SWAPPED	red	26 AWG	170	
52	DRV#3 AOUT1	Foot N20 + (red)	red	26 AWG	255	out the skin hole, 60 mm loop round the hub, through the pole channel
53	DRV#3 AOUT2	Foot N20 - (black)	black	26 AWG	255	silicone wire, it flexes every deploy

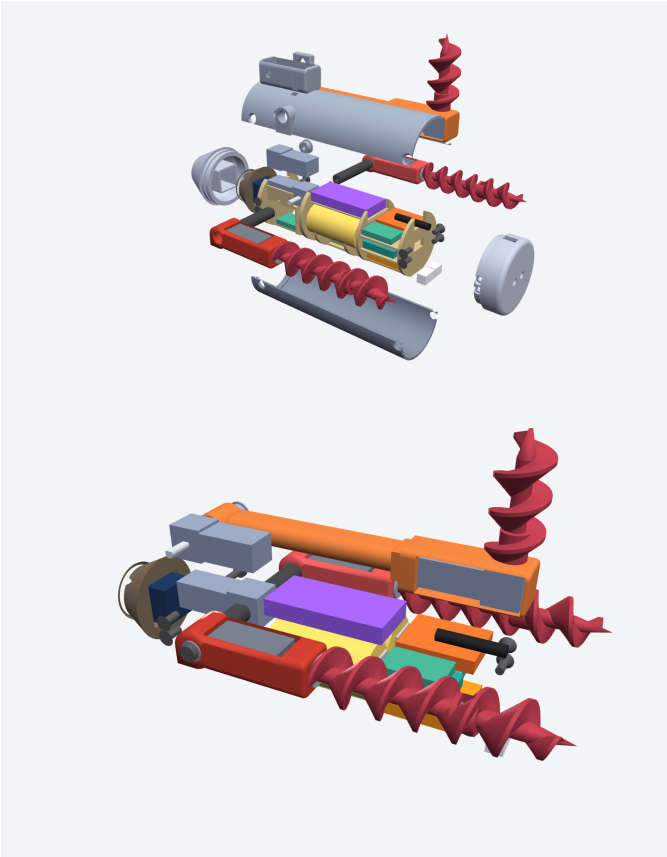
Totals by gauge and colour (mm): 22 AWG black 515, 22 AWG red 675, 24 AWG orange 160, 24 AWG black 160, 26 AWG white 180, 26 AWG black 1110, 26 AWG red 1015, 26 AWG orange 95, 28 AWG orange 105, 28 AWG black 225, 28 AWG yellow 105, 28 AWG red/white 395, 28 AWG blue 120, 28 AWG green 120, 28 AWG purple 300, 28 AWG grey 300

11. Pictures

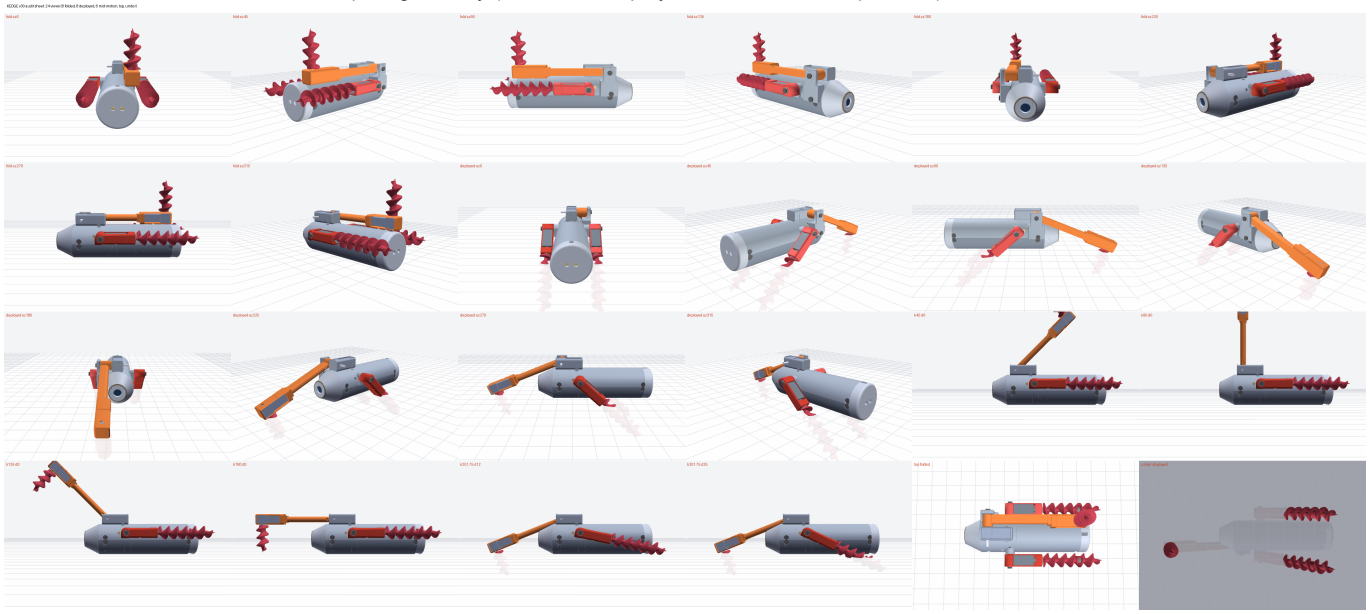
Wiring (numbers match the cut list):



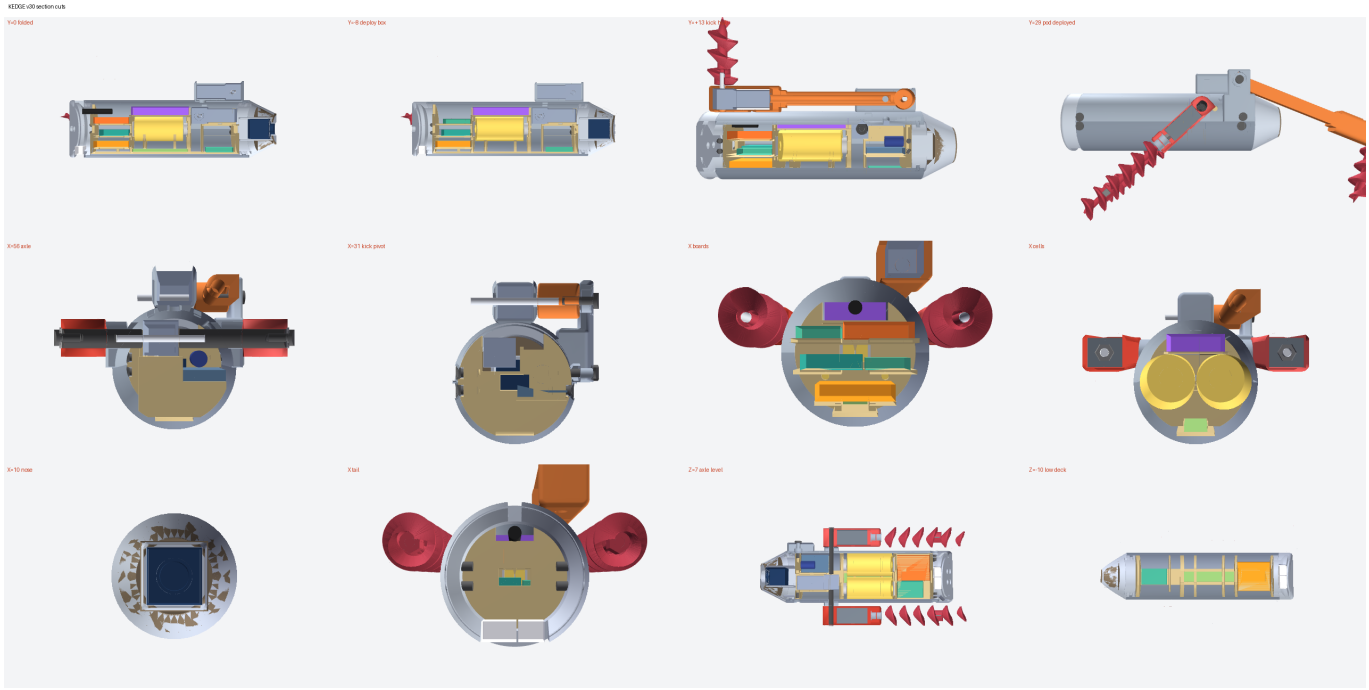
Exploded view and populated tray:



Audit sheet: 24 views of the exact print geometry (8 folded, 8 deployed, 6 mid-motion, top, under).



Section cuts:



12. Design notes

• Pod thrust: the M3 nut against the inner bearing takes the auger's pull into the ground; the auger base against the outer bearing takes push. The gearbox only carries torque.
• The drill pair counter-rotates so the dart does not spin; the two augers are mirror images so both screw in.
• The deploy N20's own shaft stubs drive the pods through two printed Ø7 D couplers that run in the hull bosses.
• Tether load path: tail end wall, tail cap, 4 radial M3 into the hull. Tie the line through the two end holes round the web.
• Ground geometry: kick stops at 201.75° with the foot housing on the ground and the foot auger 35.5 mm in; drills stop at 42.25° with the pods 0.52 mm clear and the augers 44.8 mm in.
• Power: 2S straight to the DRV8833 motor rail through the INA219; one MP1584 at 5.04 V for the XIAO, VTX and camera.
• Below the centreline the hull is a plain Ø42 cylinder for the launcher cradle; the folded pods sit just above it.